SDC HACKATHON 2026

# TECHNICAL TOOLS DECLARATION

Complete disclosure of architecture, services, models, data and responsible use

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|---|---|
| **TEAM** | Fishtail |
| **PROJECT** | FutureShift AI |
| **DOMAIN** | Domain 1 - Aiversecity Future-Ready Workforce |
| **LEADER** | YAO Yunshu |
| **UNIVERSITY** | Xiamen University Malaysia |
| **SUBMISSION FILE** | SDC2026_Fishtail_TechDeclaration.pdf |

Submission deadline: 25 October 2026

Prepared 24 October 2026

# 1. Team and Project Overview

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|---|---|
| **Team Name** | Fishtail |
| **Project / Solution** | FutureShift AI |
| **Name (Leader)** | YAO Yunshu |
| **Email & Phone** | yao18268483040@gmail.com \| +60 11-2596 4690 |
| **University** | Xiamen University Malaysia |
| **Live Prototype URL** | https://futureshift-ai.replit.app |
| **Code Repository URL** | https://github.com/SeraYao/futureshift-ai |

# 2. Architecture & Technology Stack

## 2.1 System Architecture Overview

Users enter a questionnaire or synthetic profile in a multilingual Streamlit interface. The optional CV route accepts a small text-based PDF, extracts selectable text with pypdf, applies heuristic redaction and sends at most 8,000 characters to a RunPod OpenAI-compatible endpoint only after explicit consent. Qwen/Qwen3-8B returns candidate questionnaire fields; schema validation and user review occur before profile changes. A deterministic Python engine scores the approved profile against a local, quality-controlled O*NET catalogue. BLS projections, local SDG mappings and learning resources remain separate evidence layers. Career selection feeds a four-week gap-based plan, progress tracking and JSON export/import. Streamlit session state holds user inputs; versioned JSON files hold public catalogues. There is no account database. Replit hosts the public application and stores secrets as environment variables. The questionnaire, recommendations, SDG cards and plans remain available when the optional AI endpoint fails.

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|---|---|
| **1. USER** | Questionnaire or synthetic profile; optional text-based PDF CV |
| | ↓ |
| **2. STREAMLIT APP** | Consent, validation, session state, multilingual UI |
| | ↓ |
| **3. DETERMINISTIC ENGINE** | O*NET fit, evidence display, SDG cards, four-week plan |
| | ↓ |
| **4. OPTIONAL AI** | Redacted CV text -> RunPod OpenAI-compatible API -> Qwen/Qwen3-8B |
| | ↓ |
| **5. DEPLOYMENT** | Replit public app; local JSON evidence catalogues; no account database |

## 2.2 Languages, Frameworks & Libraries

| Name & Version | Purpose | License |
|---|---|---|
| Python 3.11+ (3.14.8 local) | Application and deterministic business logic | PSF License |
| Streamlit 1.66.0 | Web UI, uploads and session state | Apache-2.0 |
| pypdf 6.21.1 | Selectable-text PDF extraction | BSD-3-Clause |

| Name & Version | Purpose | License |
|---|---|---|
| Python standard library | HTTP/TLS, JSON, files, validation | Python distribution terms |

# 3. APIs & Cloud Services

## 3.1 External APIs

| API and Provider | Purpose | Access Tier |
|---|---|---|
| RunPod OpenAI-compatible Chat Completions | Optional consent-based extraction of CV fields | Sponsor-provided / account tier to confirm |
| GitHub Search API | Isolated synthetic technology-activity demonstration only; not official career ranking | Anonymous/free; rate limited |
| BLS public HTML pages | Manual/server refresh source for US 2025-35 employment evidence | Public; no key |

## 3.2 Cloud Platforms & Deployment

| Service & Provider | Purpose | Plan / Region |
|---|---|---|
| Replit | Public Streamlit hosting and environment secrets | SDC redemption/credits; region not confirmed |
| RunPod | Hosted model inference endpoint | Sponsor-provided/account credits; region not confirmed |
| GitHub | Public source-code repository | Public repository / free tier |

No database, authentication provider, payment service, messaging API or object-storage service is used by the prototype.

# 4. AI Tools, Models & Datasets

## 4.1 AI Models Used in the Solution

| Model & Version | Source / Provider | Usage | License / Terms |
|---|---|---|---|
| Qwen/Qwen3-8B | Qwen model served via RunPod | Pre-trained; prompt-only CV field extraction; no fine-tuning | Qwen model terms and RunPod service terms apply |

The model does not score users, rank careers, generate the local SDG cards or calculate learning plans.

## 4.2 Datasets

| Dataset | Source | Type | License | Personal? |
|---|---|---|---|---|
| O*NET 31.0 | onetcenter.org | Public | CC BY 4.0 | No |
| BLS Employment Projections 2025-35 | bls.gov | Public | US government publication; attributed | No |
| UN SDG targets + project mappings | sdgs.un.org + project analysis | Public + self-created | Source attributed; no endorsement implied | No |

| Dataset | Source | Type | License | Personal? |
|---|---|---|---|---|
| Learning resource catalogue | MIT OCW, Microsoft Learn, OpenLearn, EPA, OpenStax, Purdue OWL and others | Public metadata | Provider terms; linked, not redistributed | No |
| Synthetic demo profiles, roles and trends | Created by Fishtail | Synthetic | Project Apache-2.0 | No |
| User-entered profile / optional CV text | User provided at runtime | Private, session-only | User supplied | Yes |

## 4.3 AI-Assisted Development Tools

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|---|---|
| **Generative-AI tools used?** | Yes |

| Tool | Used For | Extent |
|---|---|---|
| OpenAI ChatGPT / Codex | Code drafting and review, debugging, tests, documentation, submission preparation | Major |
| Replit Agent | Deployment configuration and troubleshooting in the Replit environment | Moderate |
| Qwen/Qwen3-8B | Runtime CV extraction testing with synthetic or non-sensitive text | Minor development validation |

### Review responsibility

The team leader reviewed generated changes against the repository, ran automated regression tests, checked JSON and Python syntax, exercised the main browser journey, and verified multilingual state, plan import/export and failure handling. Source claims were checked against official O*NET, BLS, UN and resource pages. AI output was treated as draft material; unsupported claims were removed, deterministic recommendation logic was kept separate from model output, and the team leader accepts responsibility for the final code and documents.

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# 5. Sponsor-Provided Resources

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|---|---|
| **Official sponsor resource used?** | Yes |

| Sponsor | Resource | How Used |
|---|---|---|
| Replit | SDC redemption code / deployment credits | Hosted the public Streamlit prototype |
| RunPod | Inference access / credits | Served Qwen/Qwen3-8B through an OpenAI-compatible endpoint for optional CV extraction |

### Replaceability

Replit is required for the submitted public URL but the Streamlit application can run on another Python host. RunPod is optional to the main questionnaire-to-plan workflow and is replaceable by another OpenAI-compatible provider after changing environment variables. If the model is unavailable, users can complete the questionnaire manually. Local occupational, SDG and learning catalogues remain functional without either service.

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# 6. Security, Privacy & Responsible Use

## 6.1 Data Handling & Privacy

Questionnaire answers, career choices, plans and progress are held in the user's Streamlit session; the application has no account database or server-side personal-profile store. Uploaded PDFs are parsed in application memory and are not intentionally written to project disk or logs. Only after explicit consent, heuristically redacted CV text (maximum 8,000 characters) and extraction instructions are sent to the configured RunPod model endpoint. PDF bytes, original filename, recommendation scores and career catalogue are not included in that request. Extracted candidate fields and short evidence snippets remain in the session for review. Users keep any downloaded plan files. The project does not define the cloud platform's memory cleanup timing or RunPod

log-retention/deletion SLA; therefore users are warned that redaction cannot guarantee anonymity and should use the questionnaire for sensitive data.

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## 6.2 Known Limitations & Risks

Qwen extraction can omit or misclassify fields, time out or return invalid structure; schema validation, explicit review, retry and a manual questionnaire fallback mitigate these risks. The deployed PDF upload has previously returned HTTP 403 in Replit, so the public demo should rely on the questionnaire until upload is revalidated. Scanned PDFs are unsupported because there is no OCR. The six-career official catalogue is fixed and not a live crawl. O*NET coverage and suppressed fields limit comparability. BLS data is a US 2025-35 forecast, not Malaysian demand, and is separated from personal fit. SDG links are project interpretations, not UN endorsements. The four-week plan supports practice but does not confer qualifications. GitHub activity is an isolated proxy demonstration and is not presented as recruitment demand. Provider availability, rate limits and changing public pages may interrupt refreshes; validated snapshots and clear dates are retained.

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# 7. Compliance & Readiness

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|---|---|
| **[x]** | All APIs, cloud services, AI tools, models and datasets used in the project are declared. |
| **[x]** | Third-party code, datasets and models are attributed and used under their applicable licenses or terms. |
| **[x]** | No API keys, passwords or other credentials are exposed in the repository, prototype or this document. |
| **[x]** | Generative-AI use for development is disclosed and was reviewed by the team leader. |
| **[x]** | Official SDC sponsor resources are declared in Section 5. |
| **[x]** | The prototype is accessible to judges through the live and repository links in Section 1. |
| **[x]** | All fields are completed within the stated word limits. |

# 8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model and sponsor resource used in the project has been declared.

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|---|---|
| **Team Leader** | YAO Yunshu |
| **Signature** | _____ |
| **Date** | 24 October 2026 |

Final action before upload: sign the line above and confirm the exact Replit and RunPod plan/region labels from the account dashboards. No credentials should be added.

## Reference URLs

O*NET Database | BLS Employment Projections | UN SDGs | Replit Documentation | RunPod Documentation